

Unit 11, Lesson 5: Is it Proportional?

Benchmarks

MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph, or written description.

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

Learning Targets

- ☐ I can determine if two quantities have a proportional relationship.
- ☐ I can determine the constant of proportionality of a proportional relationship.

11.5.1 Warm-Up: Visual Pattern

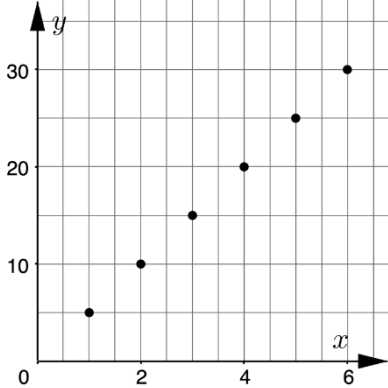
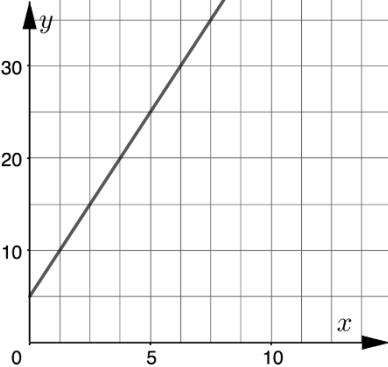
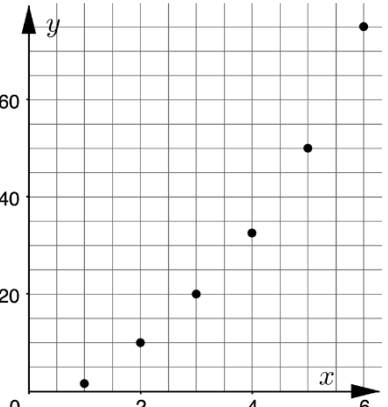
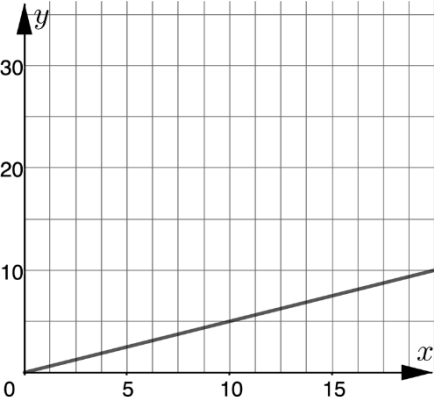
1. The first 3 steps in a pattern created using Froot Loops are shown.



- a. Describe how the pattern is changing in each step.
- b. Draw what you think the 6th step in this pattern will look like using small circles to represent each Froot Loop.

11.5.2 Guided Practice: Determine if a Relationship is Proportional from Tables and Graphs

1. Four graphs are shown in the chart. Indicate whether each graph represents a proportional relationship.

Graph A	Graph B
 ☐ proportional ☐ not proportional	 ☐ proportional ☐ not proportional
Graph C	Graph D
 ☐ proportional ☐ not proportional	 ☐ proportional ☐ not proportional

For a proportional relationship, the constant of proportionality can be determined from a graph in two ways.

- Use the point $(1, k)$, where k is the constant of proportionality.
- Find the value of the ratio $\frac{y}{x}$ using any point on the graph.

2. Two of the graphs in the previous question represent proportional relationships. Complete the statements to state the value of the ratio between the two quantities in each of those two graphs.
- Graph _____ represents a proportional relationship where the constant of proportionality is _____.
 - Graph _____ represents a proportional relationship where the constant of proportionality is _____.
3. Dylan and Maya were running laps around the football field at Hernando High School. The coach recorded their times at the end of laps 2, 4, 6, and 8.
- Complete the table to find the ratio of minutes to laps for Dylan throughout his run.

Dylan's Run		
Distance (laps)	Time (minutes)	Minutes per Lap
2	4	
4	10	
6	15	
8	23	

- Complete the table to find the ratio of minutes to laps for Maya throughout her run.

Maya's Run		
Distance (laps)	Time (minutes)	Minutes per Lap
2	5	
4	10	
6	15	
8	20	

- Complete the statement.

The table of values representing
☐ Dylan's
☐ Maya's
 run is a proportional relationship. The constant of proportionality is _____.



When determining whether two quantities have a proportional relationship from a table, **check every pair of values** in the table to determine if the ratios between them are all equivalent.

11.5.3 Guided Instruction: Determining if a Relationship is Proportional from a Written Description

To determine if a relationship is proportional from a written description, identify pairs of values that are part of the relationship based on the information given. It is helpful to consider at least 3 or 4 pairs of values, if possible.

1. The cooking time needed when using a pressure cooker changes depending on the elevation above sea level. Cooking time should be increased by 5 percent for every 1,000 feet (ft) after 2,000 ft above sea level.
 - a. By what percentage should the cooking time be increased when cooking at an elevation of 1,000 ft?
 - b. By what percentage should the cooking time be increased when cooking at an elevation of 2,000 ft?
 - c. By what percentage should the cooking time be increased when cooking at an elevation of 3,000 ft?
 - d. Write the ratio of percentage increase to elevation in cooking time from Parts A-C.
 - e. Find the value of each ratio using a calculator.

- f. Explain whether the relationship between the elevation and cooking time is proportional.
2. The actual distance between Palm Beach, FL and Melbourne, FL is 113 miles (mi). The actual distance between Melbourne, FL and Kissimmee, FL is 55 mi. On a map of Florida, the distance between Palm Beach and Melbourne is 28.25 centimeters (cm) and the distance between Melbourne and Kissimmee is 13.75 cm.
- a. Which two quantities are being compared?
- b. Identify pairs of values of the two quantities being compared.
- c. Determine whether the relationship between the two quantities is proportional. If it is proportional, state the constant of proportionality. If it is not proportional, explain why not.

11.5.4 Try It: Determining if a Relationship is Proportional

1. The table shows the distance from which a lighthouse is visible to a boat at sea.
- a. Determine whether the table represents a proportional relationship.

Height of a Lighthouse (feet)	Distance from which it can be Seen (miles)
20	6
45	9
70	11
95	13
150	16

- b. Complete the statements.

The table ☐ does ☐ does not represent a proportional relationship.

- ☐ It does not have a constant of proportionality.
- ☐ The constant of proportionality is _____.

2. The Koreshan State Park in Estero, FL charges the following admission fees for vehicles entering the park.
- \$4 for vehicles occupied by 1 person
 - \$5 for vehicles occupied 2 – 8 people
 - \$2 additional per passenger over 8 people
- a. Identify pairs of values that are comparing the cost of admission into the park for a vehicle to the number of people in the vehicle.
- b. Determine whether the relationship between the two quantities is proportional. If it is proportional, state the constant of proportionality. If it is not proportional, explain why not.

3. Aluminum cans can be recycled instead of being thrown in the garbage. Some states pay people for returning cans so the aluminum can be reused for other products. The weight of 10 aluminum cans is 0.16 kilograms (kg). The weight of 25 aluminum cans is 0.4 kg.
- a. Identify pairs of values of the two quantities being compared.

 - b. Determine whether the relationship between the two quantities is proportional. If it is proportional, state the constant of proportionality. If it is not proportional, explain why not.

11.5.5 Wrap-Up: Ticket Out the Door

1. The table shows how loud a sound is, in decibels (dB), depending on how far away the listener is from it, in feet (ft).

Distance to Listener (ft)	Sound Level (dB)
5	85
10	79
20	73
40	67

Use the table to complete the statements.

The table

does

does not

 represent a proportional relationship.

- It does not have a constant of proportionality.

The constant of proportionality is _____.

